IGCSE 0607 Practice Questions — E10.4

Averages and measures of spread

Total Questions: 15 — **Total Marks:** 105 Marks

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1 A student records the number of hours she spends reading each week for seven weeks.
The results are:

7 4 8 4 12 9 5

(a) **(a)** Write down the mode.

..... **[1]**

(b) Find the median.

..... **[1]**

(c) Work out the range.

..... **[1]**

2 The number of goals scored by a football team in each of their last 10 matches is shown below.

2 0 1 4 1 0 3 1 2 1

(a) Find the mode.

..... **[1]**

(b) Find the median.

..... **[2]**

(c) Calculate the mean.

..... **[2]**

3 A student records the number of points scored by a basketball team in 11 matches.

14 20 8 24 16 5 26 13 22 10 18

(a) Find the median.

..... **[1]**

(b) Find the range.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

(d) Calculate the mean.

..... **[2]**

4 The list shows the number of points scored by a basketball player in 7 games.

$$7 \quad 12 \quad 8 \quad 7 \quad 12 \quad x \quad 10$$

The mode of the number of points is 7.

(a) Write down the value of x.

$x =$ ..... **[1]**

(b) Find the range of the number of points.

..... **[1]**

(c) Find the median.

..... **[1]**

(d) Calculate the mean.

..... **[2]**

(e) In the 8th game, the player scores y points.
The new mean number of points for all 8 games is 9.5.
Find the value of y.

$y =$ ..... **[2]**

5 The waiting times, in minutes, for 11 patients at a dental clinic are listed below.

15 11 23 8 50 14 18 10 15 21 13

(a) Write down the mode.

..... minutes **[1]**

(b) Find

(i) the median,

..... minutes **[2]**

(ii) the interquartile range.

..... minutes **[2]**

(c) Calculate the mean waiting time.

..... minutes **[2]**

(d) Explain why the median is a better measure of average than the mean for these data.

..... **[1]**

6 A botanist measures the heights, h cm, of 50 plants in a greenhouse.
The grouped frequency table shows the results, where x and y are positive integers.

Height (h cm)	Frequency
$0 < h \leq 10$	4
$10 < h \leq 20$	x
$20 < h \leq 30$	18
$30 < h \leq 40$	y
$40 < h \leq 50$	6

(a) Use the total number of plants to show that $x + y = 22$.

[1]

(b) The estimated mean height of the 50 plants is 27 cm.

(i) Show that $3x + 7y = 122$.

[3]

(ii) Solve the simultaneous equations $x + y = 22$ and $3x + 7y = 122$ to find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(c) Write down the modal class of the distribution.

..... **[1]**

(d) Ten more plants are measured, and all ten have heights in the interval $30 < h \leq 40$.

(i) State the new modal class.

..... **[1]**

(ii) Calculate the new estimated mean height of all 60 plants, giving your answer as a mixed number.

..... cm **[3]**

7 The table shows the heights, h cm, of 80 plants.

Height (h cm)	**Frequency**
$10 < h \leq 20$	12
$20 < h \leq 30$	25
$30 < h \leq 40$	28
$40 < h \leq 50$	11
$50 < h \leq 60$	4

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean height.

..... cm **[4]**

8 The table shows information about the heights, h cm, of 50 plants.

Height (h cm)	$0 < h \leq 10$	$10 < h \leq 20$	$20 < h \leq 30$	$30 < h \leq 40$	$40 < h \leq 50$
Frequency	4	12	20	10	4

(a) **(a)** Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean height.

..... cm **[4]**

9 A survey was conducted on the masses, m grams, of 80 apples. The results are shown in the frequency table.

Mass (m grams)	$80 < m \leq 90$	$90 < m \leq 100$	$100 < m \leq 110$	$110 < m \leq 120$	$120 < m \leq 130$
Frequency	8	22	31	14	5

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean mass of the apples.

..... g **[4]**

(c) Explain why your answer to part (b) is an estimate.

.....

..... **[2]**

10 The table shows information about the time, t seconds, taken by 100 students to complete a puzzle.

Time (t seconds)	Frequency
$10 < t \leq 20$	14
$20 < t \leq 30$	28
$30 < t \leq 40$	32
$40 < t \leq 50$	18
$50 < t \leq 60$	8

(a) Write down the modal time interval.

..... **[1]**

(b) Calculate an estimate of the mean time.

..... seconds **[4]**

(c) Calculate the percentage of students who took strictly more than 40 seconds to complete the puzzle.

..... % **[2]**

(d) Write down the interval that contains the upper quartile.

..... **[1]**

11 The table shows information about the masses, m grams, of 80 apples harvested from an orchard.

Mass (m grams)	Frequency
$100 \leq m < 120$	10
$120 \leq m < 140$	22
$140 \leq m < 160$	28
$160 \leq m < 180$	14
$180 \leq m < 200$	6

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean mass of the 80 apples.

..... g **[4]**

(c) A second crate contains 40 apples with a mean mass of 155 g. Calculate an estimate of the mean mass of all 120 apples combined.

..... g **[3]**

(d) State the class interval that contains the median mass of the 80 apples.

..... **[1]**

12 A factory produces 60 items.
The masses, m kg, of these items are recorded in the table.
The estimate of the mean mass is 48 kg.

Mass (m kg)	**Frequency**
$0 < m \leq 20$	p
$20 < m \leq 40$	14
$40 < m \leq 60$	16
$60 < m \leq 80$	q
$80 < m \leq 100$	5

(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[5]**

(b) State the class interval which contains the median mass.
You must show your working.

..... **[2]**

13 The mean of five numbers is 12.
The median of the five numbers is 11.
The mode of the five numbers is 8.
The range of the five numbers is 15.

(a) Find the five numbers.

..... **[5]**

(b) A sixth number, k, is added to the list.
The new mean is 13.
Work out the value of k.

$k =$ ..... **[2]**

14 Six numbers are given in terms of x.

$$x-3 \qquad x \qquad x+2 \qquad 2x \qquad 2x+5 \qquad 3x$$

The mean of these six numbers is 9.

(a) **(a)** Calculate the value of x.

$x =$ ..... **[3]**

(b) Find the interquartile range of the six numbers.

..... **[2]**

(c) A seventh number, y, is added to the original six numbers.
The new median of the seven numbers is 8.
Find the value of y.

$y =$ ..... **[2]**

15 The table shows information about the masses, m kg, of 50 parcels.

Mass (m kg)	Frequency
$0 < m \leq 2$	15
$2 < m \leq 4$	x
$4 < m \leq 6$	y
$6 < m \leq 8$	8
$8 < m \leq 10$	2

The estimate of the mean mass of the parcels is 3.8 kg.

(a) Use the total number of parcels to write down an equation in x and y, and show that it simplifies to $x + y = 25$.

[1]

(b) Use the estimate of the mean to show that $3x + 5y = 101$.

[4]

(c) Solve the simultaneous equations to find the value of x and the value of y.
You must show all your working.

$x =$ ..... , $y =$ ..... **[3]**

(d) Write down the interval that contains the median mass.

..... **[1]**